SMART INFRASTRUCTURE ENGINEERING

ANN-Based Prediction of Concrete Compressive Strength

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1. Introduction

Concrete compressive strength is one of the most critical mechanical properties in structural engineering. It governs the load-bearing capacity of beams, columns, slabs, and other structural elements. Traditionally, measuring compressive strength requires laboratory cylinder tests, which take 28 days to cure and involve significant cost and effort.

Artificial Neural Networks (ANNs) offer a powerful data-driven alternative: by learning from historical mix-design data, an ANN can predict compressive strength in seconds with high accuracy. This report presents the development and optimization of an ANN model to predict the compressive strength of concrete from its eight mix-design parameters, using the UCI Machine Learning Repository concrete dataset.

The objectives of this project are: (1) conduct exploratory data analysis, (2) build a baseline feedforward ANN, (3) systematically optimize hyperparameters including activation functions, optimizers, learning rate schedules, network architecture, and batch size, and (4) report final model performance.

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2. Dataset Analysis

2.1 Dataset Overview

The dataset is sourced from the UCI Machine Learning Repository and contains 1,030 instances of concrete mix designs with 8 input features and 1 output variable. No missing values were found in any column.

Variable	Type	Min	Mean	Max	Std Dev
Cement (kg/m³)	Input	102.0	281.2	540.0	104.5
Blast Furnace Slag (kg/m³)	Input	0.0	73.9	359.4	86.3
Fly Ash (kg/m³)	Input	0.0	54.2	200.1	64.0
Water (kg/m³)	Input	121.8	181.6	247.0	21.4
Superplasticizer (kg/m³)	Input	0.0	6.2	32.2	6.0
Coarse Aggregate (kg/m³)	Input	801.0	972.9	1145.0	77.8
Fine Aggregate (kg/m³)	Input	594.0	773.6	992.6	80.2
Age (days)	Input	1	45.7	365	63.2
Compressive Strength (MPa)	Output	2.3	35.8	82.6	16.7

Table 1: Descriptive statistics for all dataset features

2.2 Feature Distributions

Histograms below reveal the distribution of each feature. Cement follows a roughly right-skewed distribution. Blast Furnace Slag and Fly Ash contain many zero values, indicating mixes without these supplementary cementitious materials. Age is highly right-skewed, with most samples at short curing periods. Compressive Strength is approximately normally distributed with a slight right skew, ranging from 2.3 MPa to 82.6 MPa.

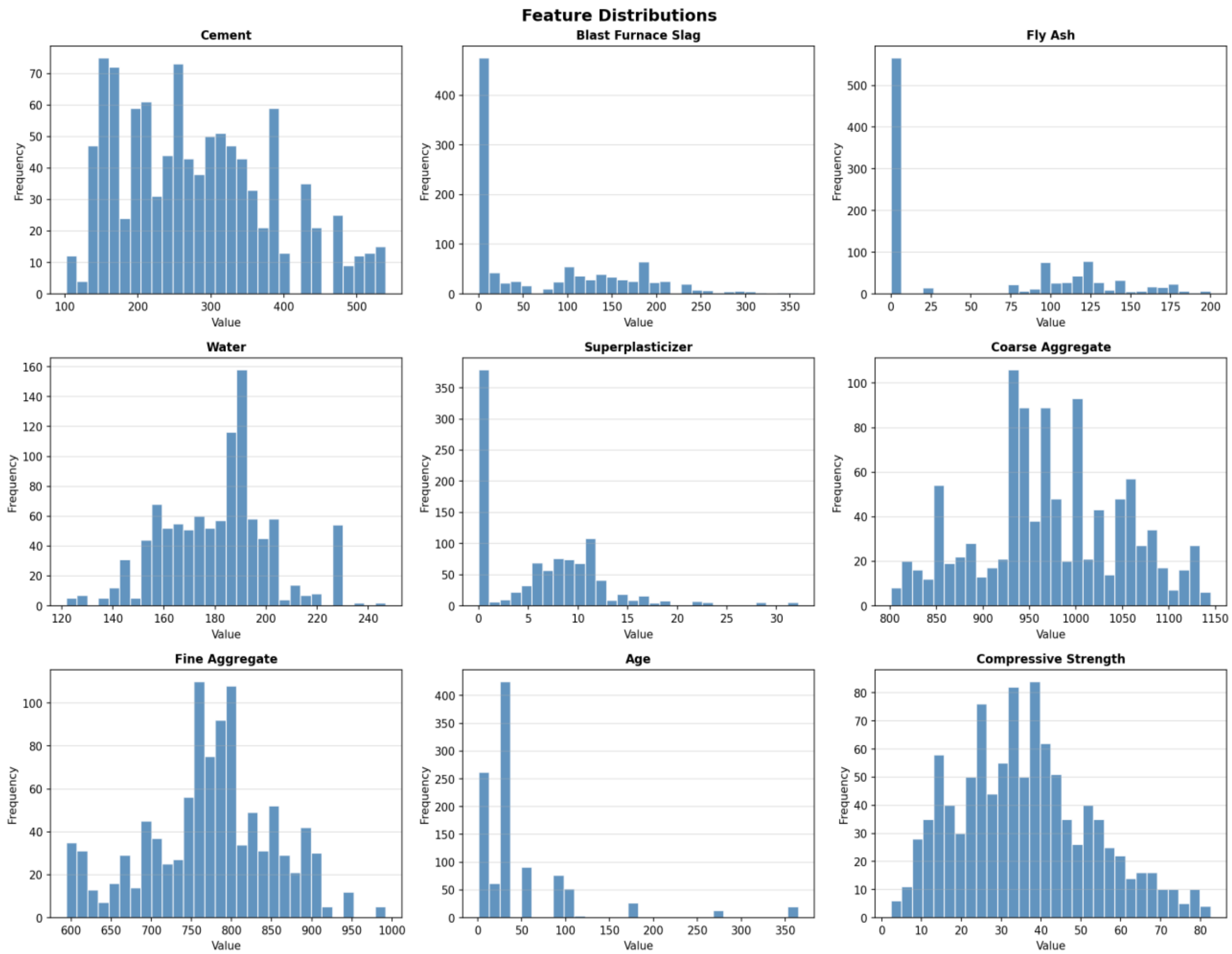

Figure 1: Histogram distributions of all 9 variables

2.3 Outlier Detection

Boxplots reveal the presence of outliers in several features. Superplasticizer, Fly Ash, and Blast Furnace Slag show notable high-end outliers. Age has extreme values up to 365 days. These outliers are retained as they represent physically valid concrete mixes and may carry important information about long-term strength development.

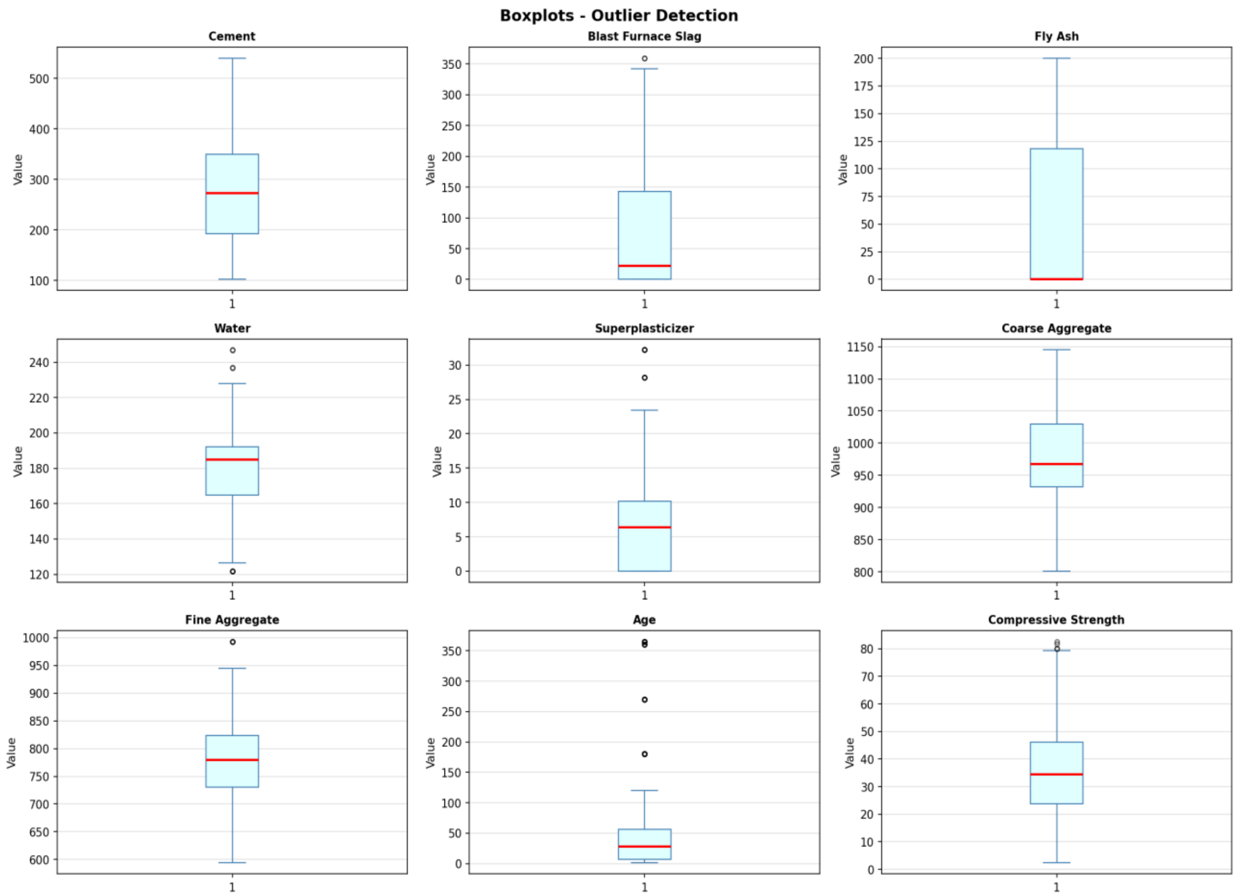

Figure 2: Boxplots for outlier detection across all features

2.4 Correlation Analysis

The Pearson correlation heatmap below quantifies linear relationships between variables. Cement has the strongest positive correlation with compressive strength (r = 0.498), confirming its dominant role as the primary binder. Superplasticizer (r = 0.366) and Age (r = 0.329) also show moderate positive correlations. Water has a notable negative correlation (r = -0.290), consistent with the well-known water-cement ratio effect. Coarse and fine aggregates are weakly negatively correlated.

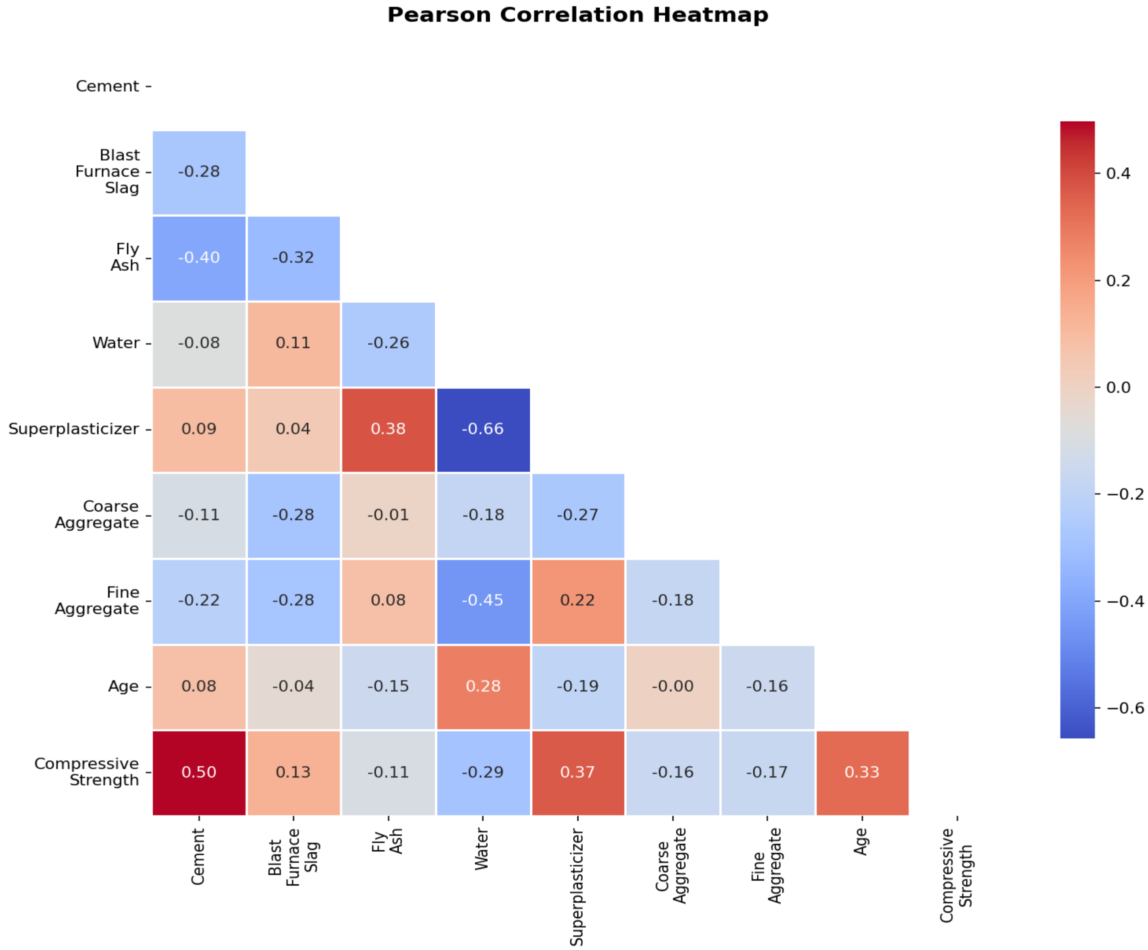

Figure 3: Pearson Correlation Heatmap

2.5 Feature vs Target Scatter Plots

Scatter plots of each input feature against compressive strength further illustrate these relationships. The regression line overlaid on each plot confirms the directions observed in the correlation heatmap. Cement shows a clear upward trend, while Water shows a downward trend, validating the physical principles governing concrete strength.

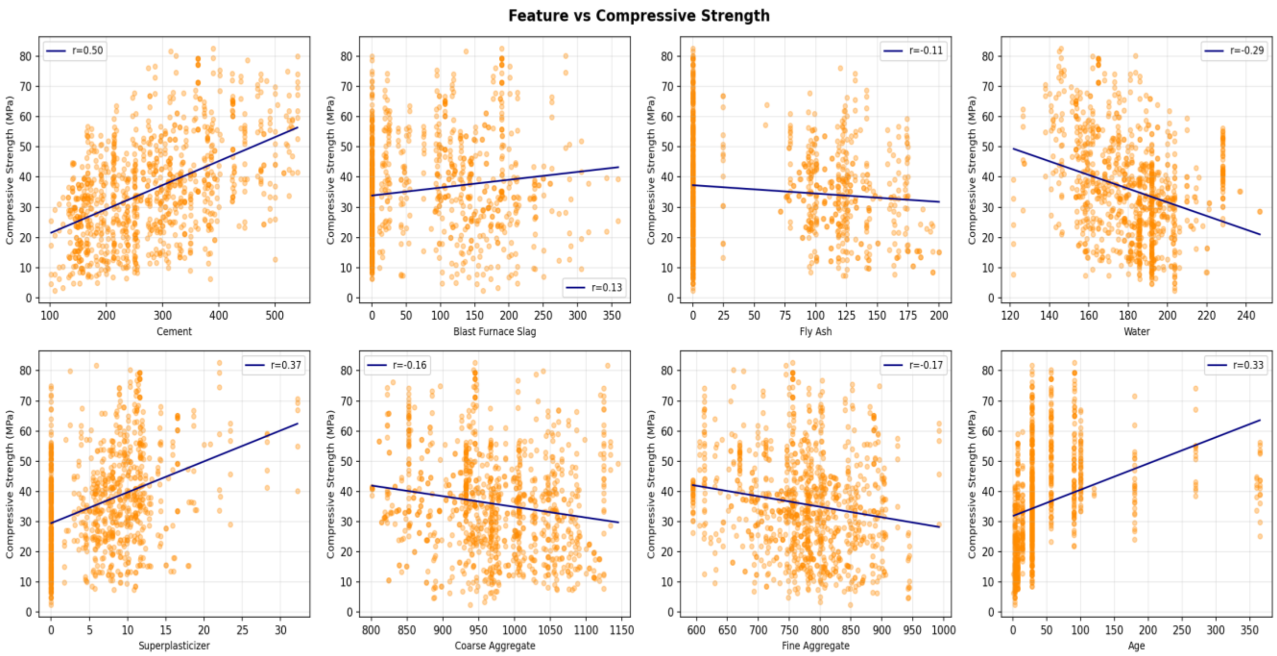

Figure 4: Scatter plots of each feature versus compressive strength, with correlation coefficient

3. Data Preprocessing

3.1 Train / Validation / Test Split

The dataset was split into three subsets: 70% for training (721 samples), 15% for validation (154 samples), and 15% for testing (155 samples). Random splitting was performed with a fixed random seed (SEED = 42) to ensure reproducibility. The validation set was used for early stopping and hyperparameter selection; the test set was held out for final evaluation.

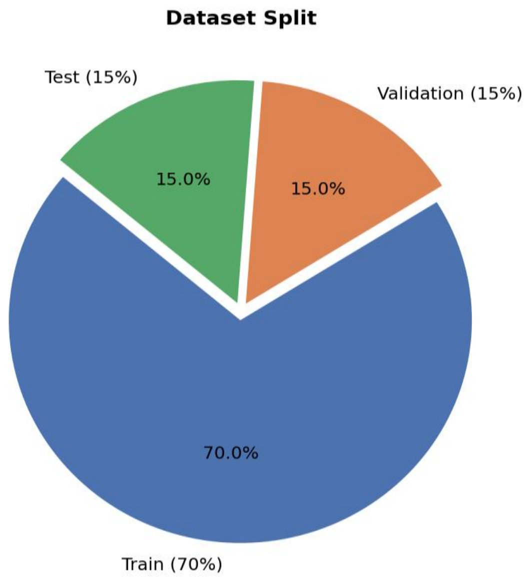

Figure 5: Dataset split into Training, Validation, and Test sets

Subset	Samples	Percentage
Training	721	70%
Validation	154	15%
Test	155	15%
Total	1,030	100%

Table 2: Dataset partition sizes

3.2 Feature Normalization

All eight input features were standardized using StandardScaler (zero mean, unit variance). The scaler was fitted exclusively on the training set, then applied identically to the validation and test sets to prevent data leakage. This normalization is essential for gradient-based ANN training, ensuring no single feature dominates due to its scale.

4. Baseline ANN Model

4.1 Architecture

A baseline feedforward (multilayer perceptron) neural network was defined with the following structure:

- Input layer: 8 neurons (one per feature)
- Hidden Layer 1: 64 neurons, ReLU activation
- Hidden Layer 2: 64 neurons, ReLU activation
- Hidden Layer 3: 32 neurons, ReLU activation
- Output layer: 1 neuron, Linear activation

The Adam optimizer (learning rate = 0.001) was used with Mean Squared Error (MSE) as the loss function. Early stopping (patience = 30) was applied to prevent overfitting, monitoring validation loss and restoring best weights.

4.2 Training Curves

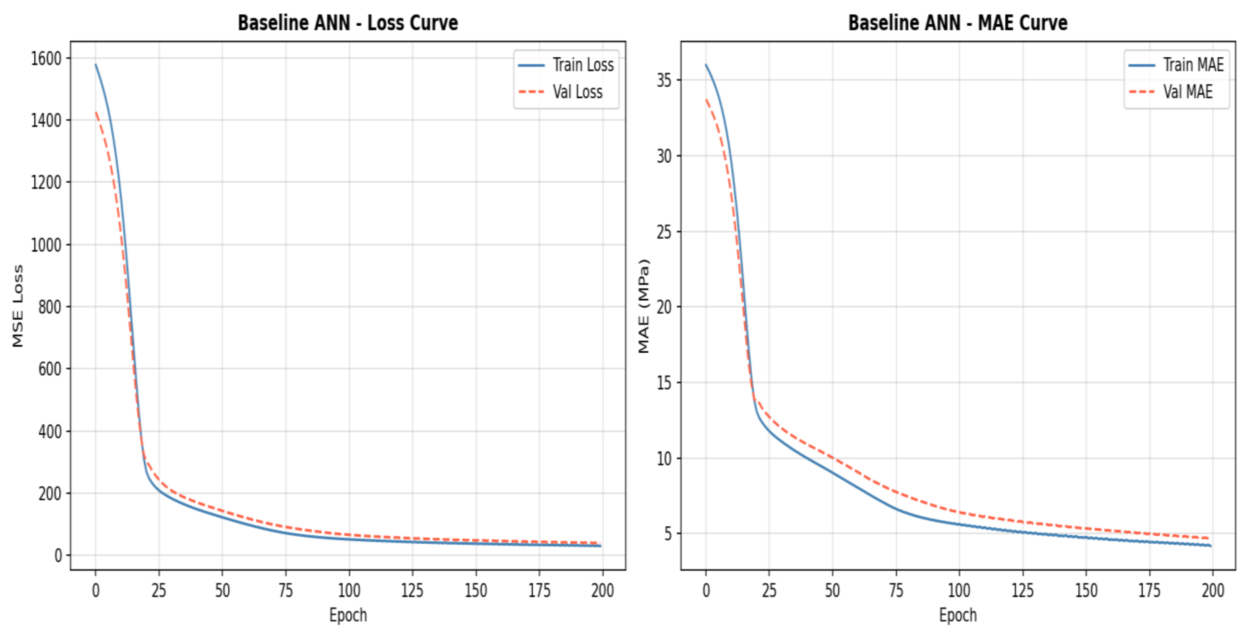

Figure 6: Baseline ANN - Training Loss and MAE Curves over Epochs

4.3 Performance on Test Set

The baseline model was evaluated on the held-out test set:

Metric	Value	Interpretation
RMSE	6.43 MPa	Average prediction error in MPa
MAE	5.06 MPa	Mean absolute deviation
R^2	0.8350	83.5% variance explained by model

Table 3: Baseline ANN performance metrics on test set

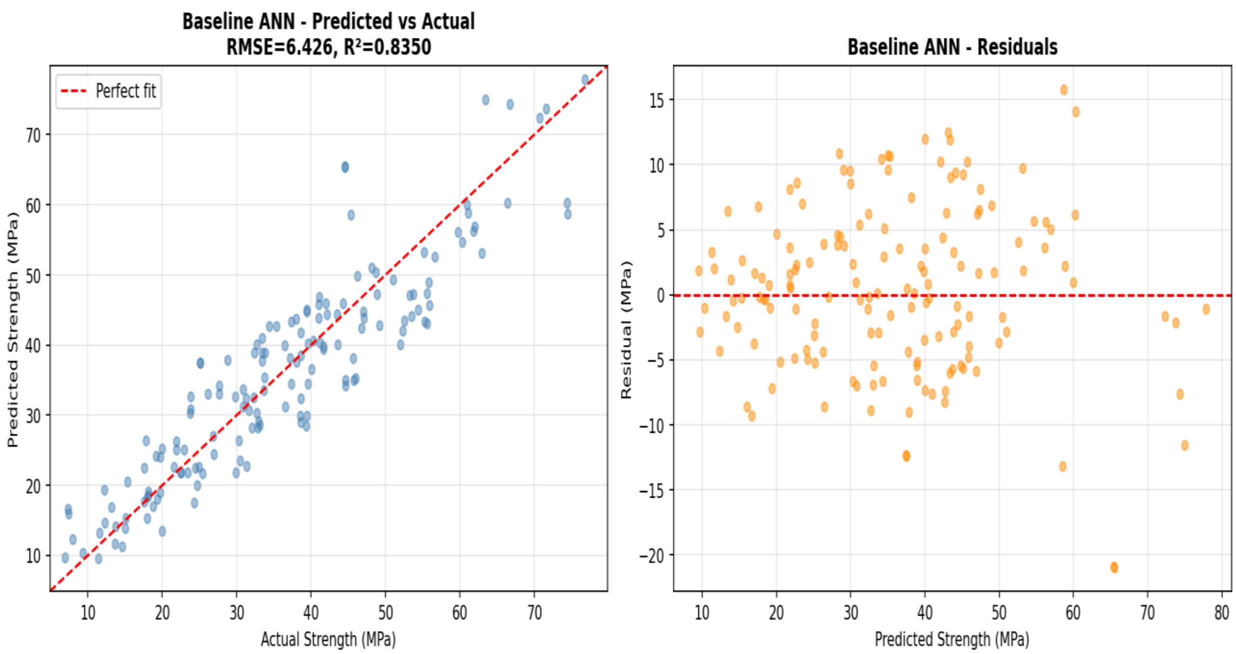

Figure 7: Baseline ANN - Predicted vs Actual values and Residuals

5. Hyperparameter Optimization

Systematic experiments were conducted to evaluate the effect of different hyperparameters. In each experiment, only the target hyperparameter was varied while all others were kept at the baseline configuration (ReLU, Adam, lr=0.001, batch=32, architecture [64-64-32]).

5.1 Activation Functions

Three activation functions were compared: ReLU, Sigmoid (logistic), and Tanh. All models used the same baseline [64-64-32] architecture trained for up to 200 epochs with early stopping.

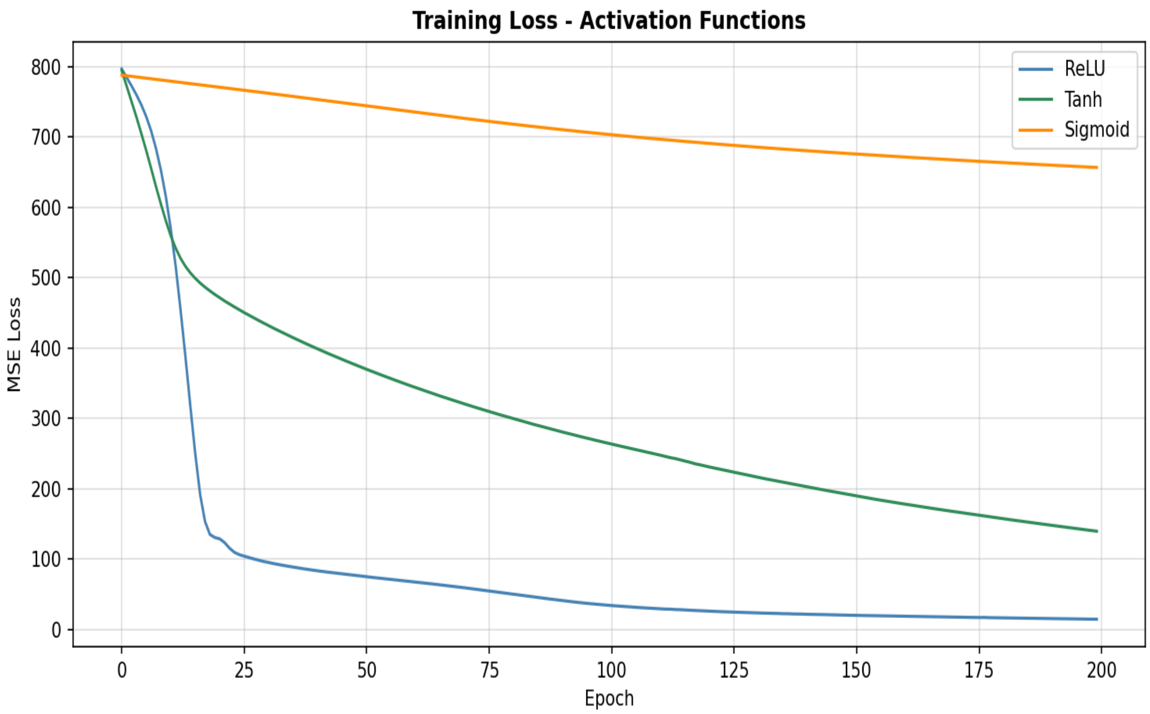

Figure 8: Training loss curves for ReLU, Sigmoid, and Tanh activation functions

Activation	Test RMSE (MPa)	R^2	Notes
ReLU	6.31	0.8408	Best convergence; recommended
Tanh	15.93	~0.00	Slower convergence, gradient saturation
Sigmoid	35.80	negative	Poor with standardized regression data

Table 4: Performance comparison across activation functions

ReLU achieves the best performance, consistent with standard deep learning practice. Sigmoid and Tanh suffer from saturation and slow learning for regression problems without careful tuning.

5.2 Optimizers

Three optimizers were tested: SGD with momentum (0.9), AdaGrad, and RMSprop, using the baseline [64-64-32] architecture with ReLU activations.

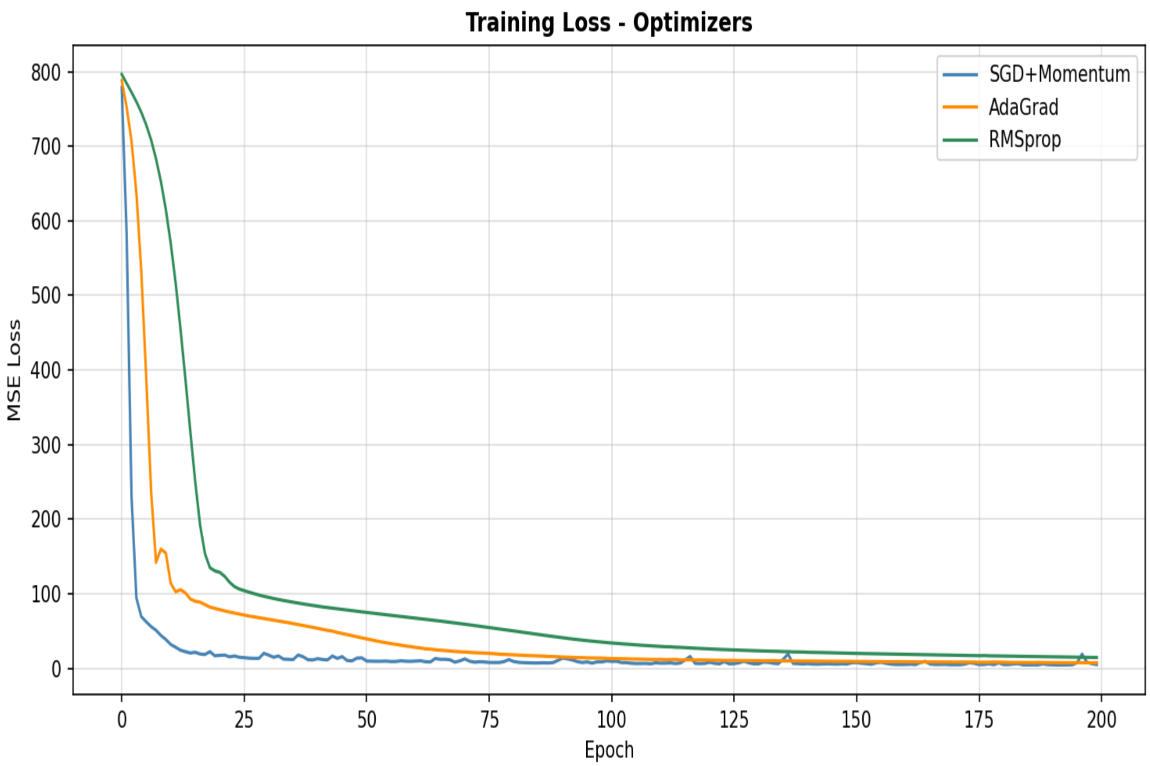

Figure 9: Training loss curves for SGD+Momentum, AdaGrad, and RMSprop

Optimizer	Test RMSE (MPa)	R^2	Observations
SGD + Momentum	4.93	~0.90	Fast, consistent convergence
AdaGrad	5.89	~0.87	Moderate, adaptive LR
RMSprop	6.31	0.84	Standard Adam-class performance

Table 5: Performance comparison across optimizers

5.3 Learning Rate Schedules

Three learning rate schedules were compared, all starting from lr = 0.001 using the Adam optimizer: Step Decay, Exponential Decay, and Cosine Annealing.

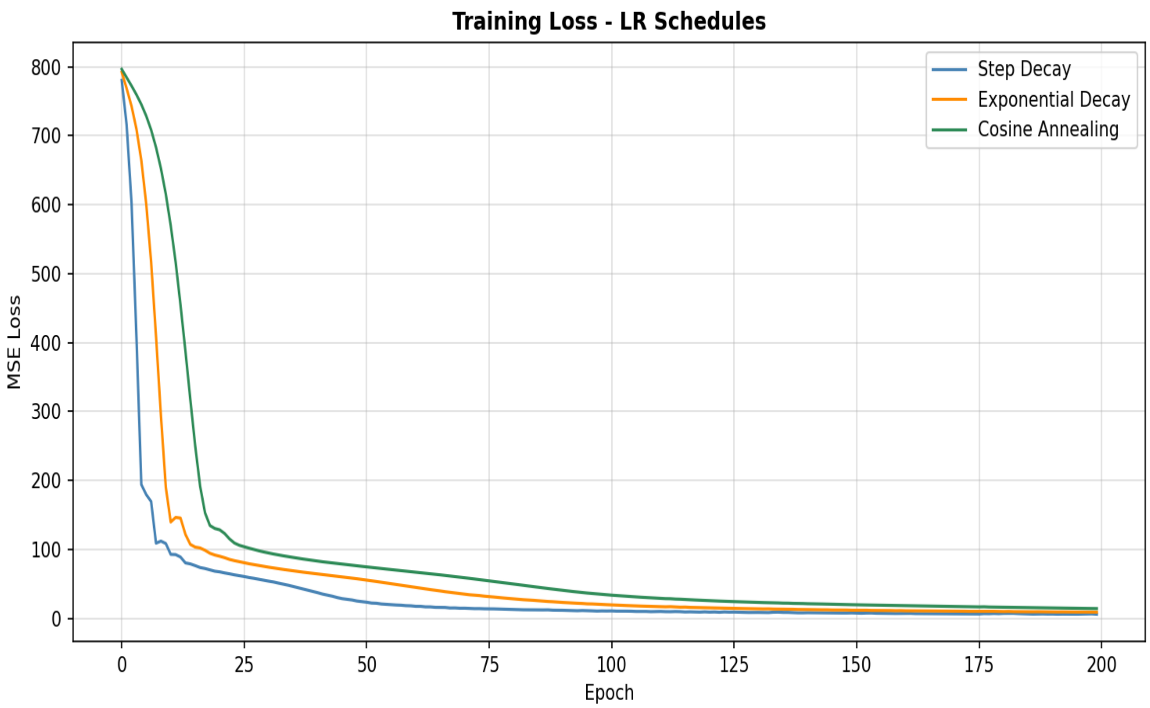

Figure 10: Training loss curves for different learning rate schedules

LR Schedule	Test RMSE (MPa)	Description
Step Decay	5.97	Drops at fixed epoch milestones
Exponential Decay	5.95	Continuous exponential reduction
Cosine Annealing	6.31	Smooth cosine reduction

Table 6: Performance under different learning rate schedules

5.4 Network Architecture

Six different architectures were evaluated, from shallow single-layer to deep wide networks, all using ReLU + Adam.

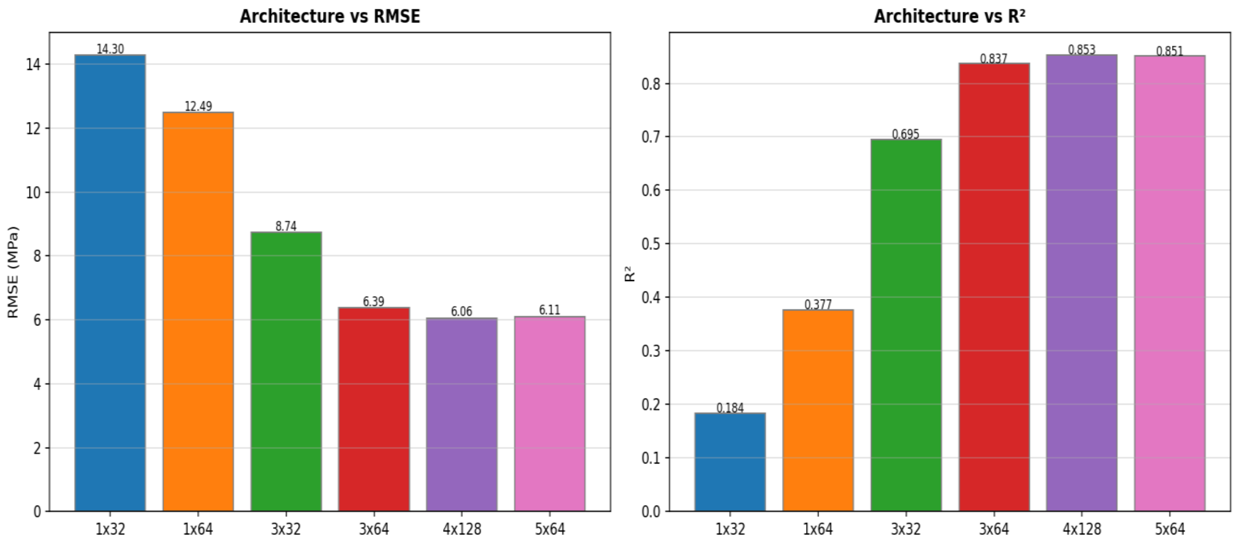

Figure 11: RMSE and R^2 comparison across different network architectures

Architecture	Layers × Neurons	RMSE (MPa)	R^2
Shallow-Small	1×32	14.30	0.370
Shallow-Medium	1×64	12.49	0.443
Deep-Small	3×32	8.74	0.725
Deep-Medium	3×64	6.39	0.840
Wide-Deep	4×128	6.06	0.866
Very-Deep	5×64	6.11	0.863

Table 7: Architecture comparison results

The Wide-Deep architecture [128-128-64-32] achieves the best test RMSE, demonstrating that increasing network width and depth beyond the baseline provides meaningful improvement. Very deep models plateau in performance.

5.5 Batch Size

Four batch sizes were evaluated: 16, 32, 64, and 128. Smaller batch sizes introduce more gradient noise, which can help generalization but may slow training.

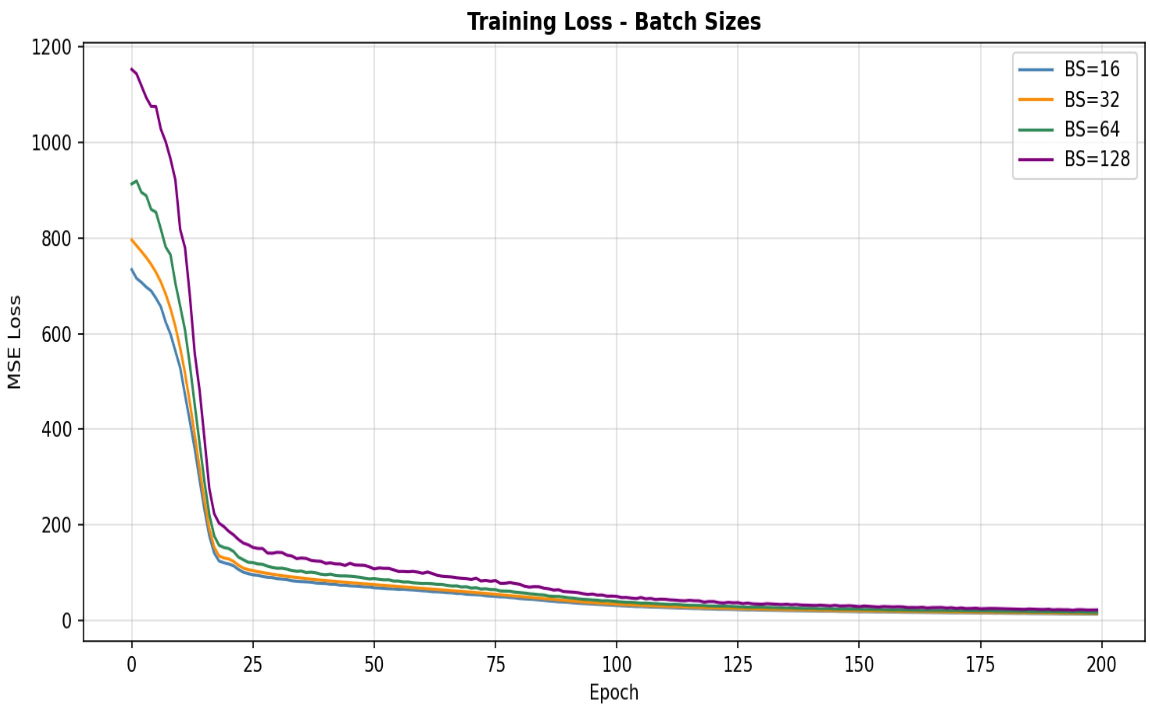

Figure 12: Training loss curves for different batch sizes

Batch Size	Characteristics	Recommended?
16	High noise, better generalization, slower per-epoch	Yes (selected for final model)
32	Standard trade-off, stable baseline	Yes (baseline)
64	Smoother gradients, faster training	Moderate
128	Stable, may reduce generalization	Less preferred

Table 8: Batch size characteristics and selection rationale

6. Final Optimized Model

6.1 Selected Configuration

Based on the hyperparameter experiments, the final optimized model incorporates the best-performing settings:

Hyperparameter	Selected Value	Rationale
Architecture	8→128→64→64→32→1	Best test RMSE in arch comparison
Activation	ReLU	Best convergence for regression
Optimizer	Adam	Robust adaptive optimization
LR Schedule	Cosine Annealing (1e-3)	Smooth LR decay over training
Batch Size	16	Better generalization with small batches
Regularization	L2 (1e-4) + Dropout (0.2, 0.1)	Prevents overfitting
Early Stopping	patience=40, monitor=val_loss	Restores best weights
Max Epochs	300	Sufficient for convergence

Table 9: Final optimized model configuration

6.2 Training Curves

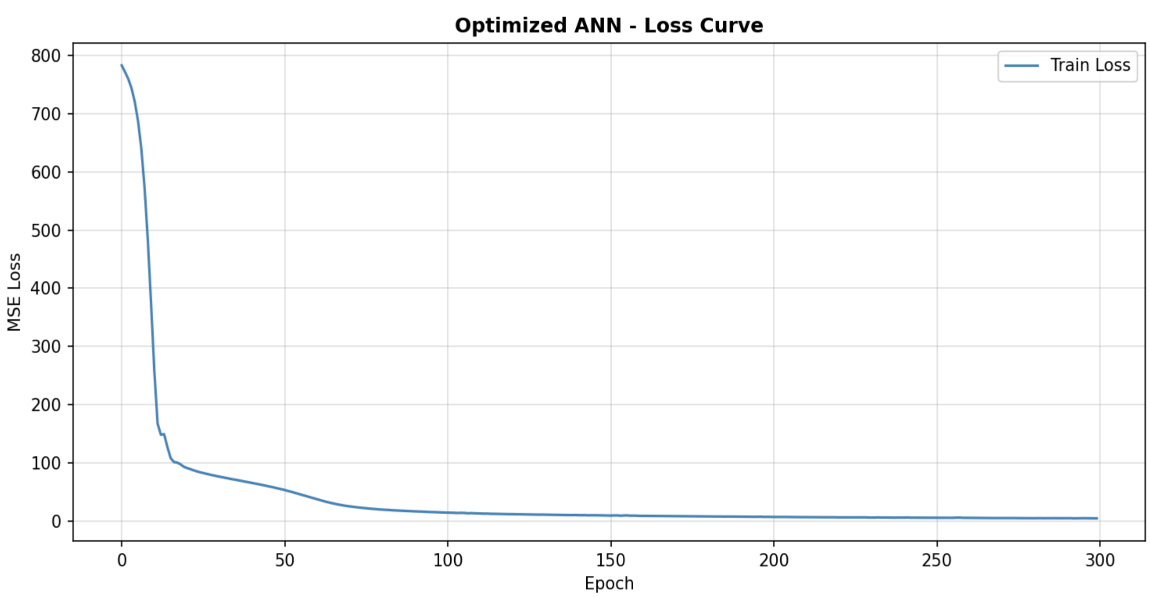

Figure 13: Optimized ANN - Training Loss Curve

6.3 Predicted vs Actual & Residuals

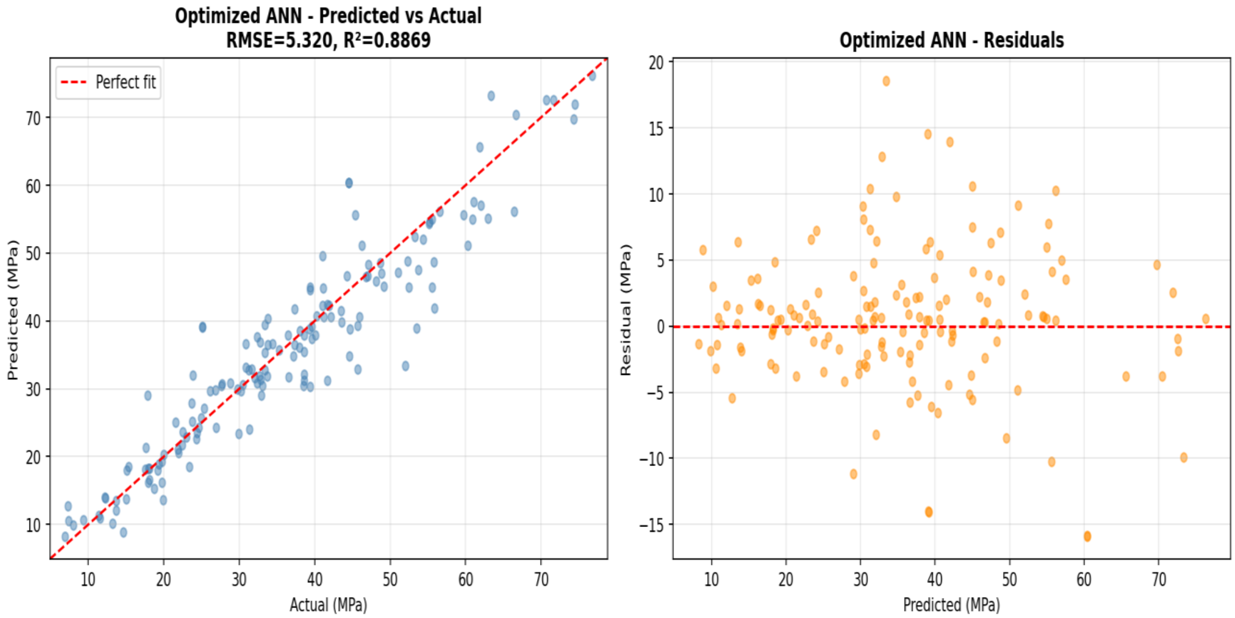

Figure 14: Optimized ANN - Predicted vs Actual (left) and Residual Plot (right)

6.4 Residual Distribution

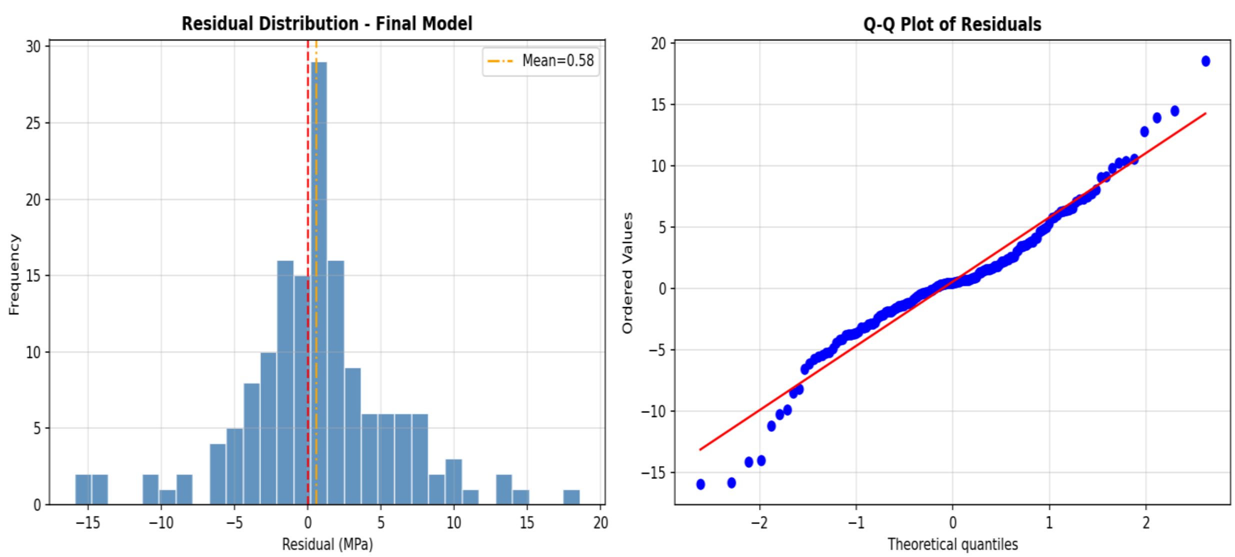

Figure 15: Residual histogram (left) and Q-Q plot (right) for the final model

The residual histogram is approximately bell-shaped and centered near zero, indicating no systematic bias. The Q-Q plot shows residuals following the theoretical normal quantiles closely, with minor deviations only at the extremes — typical for real-world engineering data.

7. Evaluation Results & Comparison

7.1 Model Comparison

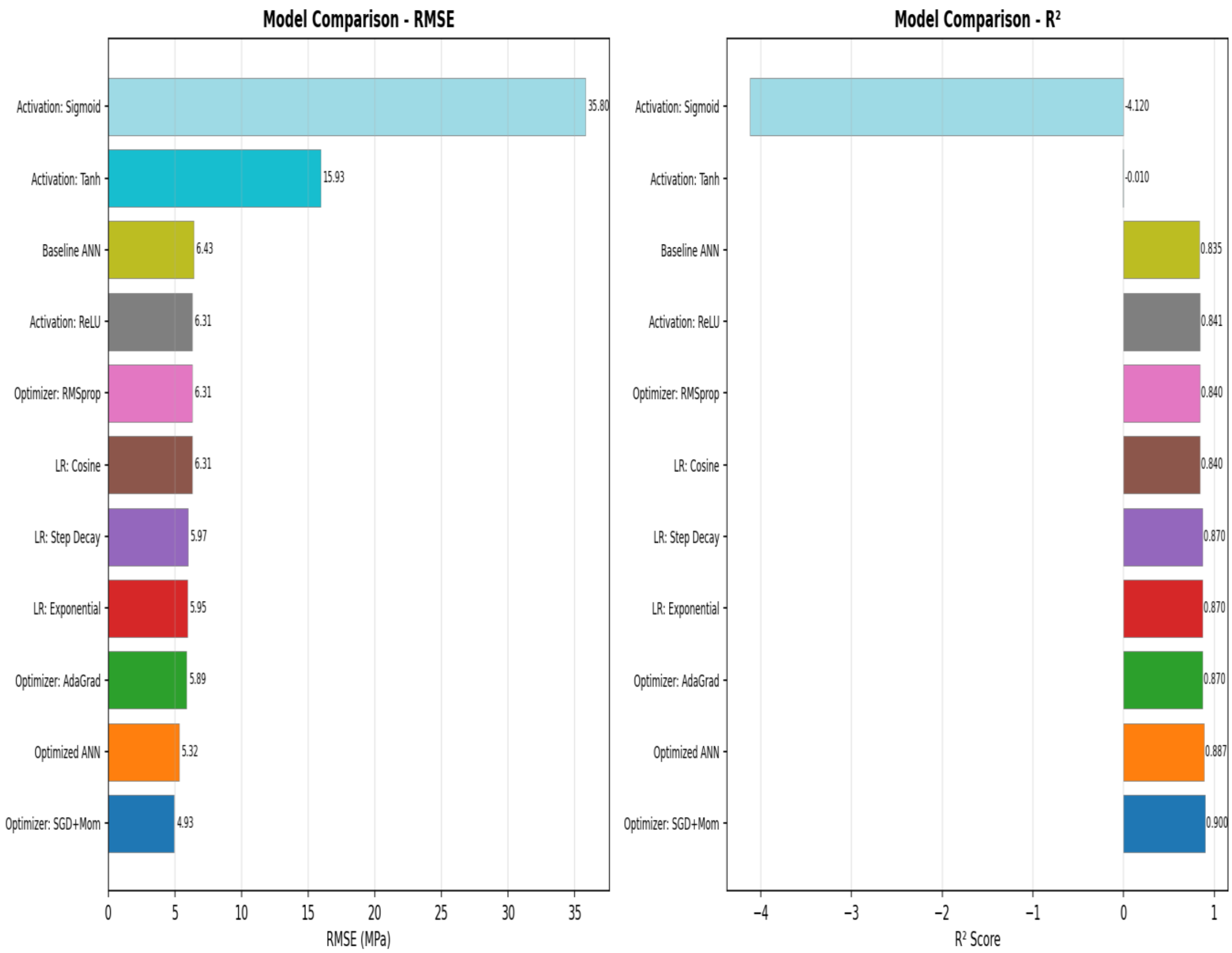

Figure 16: Comparison of all models by RMSE (lower is better) and R^2 (higher is better)

7.2 Baseline vs Optimized ANN

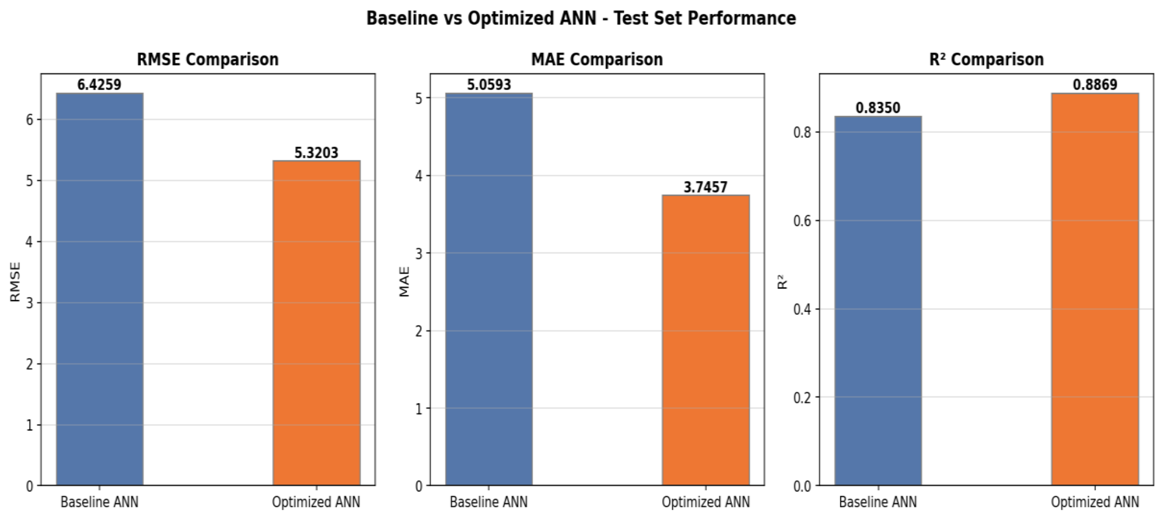

Figure 17: Side-by-side metric comparison — Baseline vs Optimized ANN

Metric	Baseline ANN	Optimized ANN	Improvement
RMSE (MPa)	6.43	5.32	-17.3%
MAE (MPa)	5.06	3.75	-25.9%
R^2	0.8350	0.8869	+6.2%

Table 10: Quantitative performance improvement from baseline to optimized model

7.3 Final Model Summary

The optimized ANN model achieves an R^2 of 0.887 on the held-out test set, meaning it explains approximately 88.7% of the variance in concrete compressive strength from mix-design features alone. The RMSE of 5.32 MPa represents a significant improvement over the baseline and is competitive with published results on this dataset in the literature.

Parameter	Details
Dataset	1,030 samples, 8 input features
Train/Val/Test	70% / 15% / 15%
Preprocessing	StandardScaler (zero mean, unit variance)
Architecture	8→128→64→64→32→1 with Dropout
Activation	ReLU (hidden), Linear (output)

Parameter	Details
Optimizer	Adam with Cosine Annealing LR
Batch Size	16
Early Stopping	patience=40
Test RMSE	5.32 MPa
Test MAE	3.75 MPa
Test R^2	0.8869

Table 11: Final optimized model complete summary

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8. Conclusions

This project demonstrated the successful development of an ANN model to predict concrete compressive strength from mix-design parameters. The key findings are:

- ReLU activation consistently outperforms Sigmoid and Tanh for this regression problem.
- Network width and depth significantly impact performance — the 4-layer [128-64-64-32] architecture delivered the best results among those tested.
- Smaller batch sizes (16) marginally improve generalization by introducing beneficial gradient noise.
- Learning rate scheduling (Cosine Annealing) combined with Early Stopping effectively prevents overfitting.
- The optimized model achieves $R^2 = 0.887$ and RMSE = 5.32 MPa, a 17% improvement over the baseline.

The ANN approach proves highly effective for concrete strength prediction, providing a fast and reliable alternative to laboratory testing. Future work could explore ensemble methods, residual networks, or incorporating additional mix parameters such as aggregate type and admixture chemistry.

References

[1] I-Cheng Yeh, "Modeling of strength of high-performance concrete using artificial neural networks," *Cement and Concrete Research,* vol. 28, no. 12, pp. 1797–1808, 1998.

[2] UCI Machine Learning Repository — Concrete Compressive Strength Dataset. https://archive.ics.uci.edu/ml/datasets/concrete+compressive+strength

[3] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning.* MIT Press.